Localized in-gap eigenstates of a 10^4-vertex disordered photonic network from direct interior Maxwell eigensolves

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Bottom-up (611 bands, validated against MPB at 64^3) and interior (133 certified pairs near eigenvalue index 5000, validated against it) Maxwell eigensolves resolve the gap-edge modes of local-self-uniform photonic networks with 10^3 and 10^4 vertices. The 10^3 gap is empty of eigenvalues; the 10^4 pseudogap holds ten certified states, five of them bulk-localized with $\xi = 1.8$–$2.5\,\mu$m $\ll L = 24.6\,\mu$m and persistent under periodic boundary wrapping. Decay lengths funnel to $\sim 2\,\mu$m at the edges; level statistics are GOE-like in the bands, Poisson-compatible in 12 levels at the lower edge. No mobility edge is claimed.

Motivation.—Local-self-uniform (LSU) networks are disordered trivalent dielectric networks that open a full three-dimensional photonic band gap without long-range order [1], in the family of amorphous-diamond [2, 3] and hyperuniform [4–9] photonic materials. Gap formation and Anderson localization [10, 11] in such correlated disordered media are linked [12, 13], and a transition from diffusion to localization near the band edge has been observed in 3-D amorphous networks [14], but the eigenmodes *inside* and at the edges of the gap at the 10^4-vertex scale have not been resolved by direct computation. Sellers *et al.* state that an N-vertex supercell has its gap between bands $N/2$ and $N/2+1$ [1], consistent with the band count of the parent gyroid/srs crystal [15], so at $N = 10^4$ the gap sits at eigenvalue index ≈ 5000; computing all lower bands bottom-up would need 1.38 TB of locked vectors at 256^3 and $O(100)$ days, hence an interior method.

System and operator.—We use the $N = 1000$ reference network of Ref. [1] (box $L = 11.44\,\mu$m, mean rod length $0.80\,\mu$m) and an $N = 10^4$ network generated with the same protocol at the same vertex density ($0.668\,\mu\mathrm{m}^{-3}$, $L = 10^{1/3} \times 11.44 = 24.647\,\mu$m). Rods are rasterized as binary voxels (no subpixel smoothing [16, 17]) in two decorations: the "elliptical" convention of the original $N = 1000$ montage (radius $0.2252\,\mu$m in a z-space stretched by 2.5; $n = 2.9275$, $\varepsilon = 8.57$; filling fraction ff $= 21.7\%$), and a "circular" decoration (radius $0.331836\,\mu$m, $n = 2.9$, $\varepsilon = 8.41$) whose radius was bisected to ff $= 22.00\%$ at 256^3 (22.01% at 192^3; 21.91% at 128^3 for $N = 1000$). Following MPB [18] we solve $\Theta \boldsymbol{H} = \nabla \times \varepsilon^{-1} \nabla \times \boldsymbol{H} = (\omega/c)^2 \boldsymbol{H}$ in the two-component transverse plane-wave basis of the supercell at $k = \Gamma$ (six 3-D FFTs per application; the two $\omega = 0$ modes removed exactly; Θ real symmetric on real fields). Eigenvalues are quoted as $\lambda = (\omega/c)^2$ in $\mu\mathrm{m}^{-2}$ and as $\nu = \omega a/2\pi c$ with $a = L/5 = 2.288\,\mu$m (density-equivalent 8-vertex cell, the same for both sizes). Band numbering is MPB's (bands 1–2 are the $\omega = 0$ pair).

Method.—(a) For $N = 1000$ we use a deflated block LOBPCG [19, 20] with MPB's transverse-projection preconditioner [18], blocks of 32 with 12 guard vectors, complex64 on the GPU with fp64 host Gram/Rayleigh–Ritz algebra (TF32 disabled), per-band relative residual $\leq 10^{-4}$, and a host-streamed locked set. On identical 64^3 grids it reproduces MPB (fp64, tolerance 10^{-7}): max $\Delta\omega/\omega = 9.0 \times 10^{-6}$ over 298 bands and 3.5×10^{-5} over 658 bands (medians 1.4×10^{-6}). The pre-registered convergence gate failed as registered: over $64^3/96^3/128^3$ the gap-edge band 500 is non-monotone with a 0.27% spread (three orders of magnitude above the solver–MPB error: rasterization-limited; the other five sampled bands converge monotonically) and the elliptical gap width runs $2.35 \to 1.93 \to 2.08\%$. (b) For $N = 10^4$ we locate the gap with the kernel polynomial method [21, 22] (KPM; 256^3, degree 12 000, 12 probes, Jackson kernel) and solve the window $\lambda \in [1.757, 2.117]$ at 192^3 by two-stage bandpass Chebyshev filtered subspace iteration [23–26]: a Jackson step-difference filter of degree 3000–4000 builds a basis of $1.5\times$ the predicted count from random starts and the same filter at degree 12 000 polishes it; candidates are extracted only by fp64 Rayleigh–Ritz on the original Θ and accepted only if their relative residual on Θ is $\leq 10^{-4}$, which by Weyl's inequality places a true eigenvalue within $10^{-4}\lambda$ of every accepted value (spurious eigenvalues are thereby excluded; eigenvectors of the two pairs closer than that are certified only as a two-dimensional span). It won a measured bake-off on a 50-band interior slice of the exact $N = 1000$ spectrum (24/50 verified within budget vs 0/50) against folded-spectrum LOBPCG [27, 28] and shift-invert PMINRES at the tested budgets; filtered Lanczos [29] was excluded on memory grounds [Supplemental Material (SM) [30]]. Its accuracy is set by re-solving that slice in the production configuration (polish degree 8000 rather than 12 000): 50/50 found, 0 ghosts, max $\Delta\lambda/\lambda = 2.4 \times 10^{-7}$ (median 6×10^{-8})—after correcting a Rayleigh-quotient normalization bug that biased every eigenvalue high by $+2.8 \times 10^{-5}$ at 128^3 and $+4.7 \times 10^{-5}$ at 192^3 ($\lambda_{\mathrm{corr}} = \lambda_{\mathrm{raw}}/\|x\|^2$, applied retroactively; SM)—and by an independent cross-check on the circular $N = 1000$ decoration (210/210 states, max 4.1×10^{-7}). Localization is quantified per mode from $u = \varepsilon|\boldsymbol{E}|^2$ by the participation fraction $p = (\sum u)^2/(N_{\rm vox} \sum u^2)$ and an envelope decay length ξ from a linear fit of $\ln u(r)$ vs the (minimum-image) distance r from the peak over $[0.10, 0.95]\,L/2$, $u \propto e^{-2r/\xi}$; a fit is "unresolved" (lower bound only) if $\xi \geq L/2$, if u decays by less than one decade, or if $r^2 < 0.7$.

Results, N = 1000.—All 611 bands (MPB bands 3–613) of the elliptical structure at 128^3 were solved in 5.5 h; the gap lies between bands 500 and 501 as pre-registered: $\lambda = 1.8830 \to 1.9628\,\mu\text{m}^{-2}$, $\Delta\nu/\nu = 2.08\%$ at 128^3 (centre $\nu = 0.505$), a spacing 10× the median of its 20 neighbours. With the circular decoration the gap is again at 500|501 and widens to $1.8276 \to 2.0225\,\mu\text{m}^{-2}$, $\Delta\nu/\nu = 5.07\%$ (26× the neighbouring spacings). Both gaps are empty of eigenvalues. Band-edge modes are localized, deep-window modes extended [Fig. 2(a)]: circular decoration $\xi = 1.47\,\mu$m (band 500, $r^2 = 0.97$, $p = 0.42\%$) and $2.30\,\mu$m (band 501, $p = 3.9\%$) vs p = 9–37% far from the gap; elliptical $\xi = 1.82$ and $2.92\,\mu$m. Only 42 of 210 circular-decoration modes have resolved ξ (142 hit the $L/2 = 5.72\,\mu$m ceiling), the quantitative reason for the $N = 10^4$ computation.

Results, $N = 10^4$.—KPM places 5010.3 ± 12.5 transverse states below $\lambda = 1.957$, i.e. 0.501 bands per vertex, consistent with the $N/2$ rule to $\pm 0.3\%$ (gap at MPB bands $\approx 5012|5013 \pm 13$). The Jackson-smoothed KPM density falls below 160 states per unit λ over $[1.864, 1.996]\,\mu\text{m}^{-2}$ ($\Delta\nu/\nu = 3.4\%$; 5%/20% criteria: $[1.885, 1.968]/[1.837, 2.022]$, a ± 0.03 systematic; certified states inside: 5, 10, 21), narrower than the 5.07% gap of the $N = 1000$ box with the same decoration and contained within it. Three overlapping slices at 192^3 (69 + 5 + 61 pairs, two duplicates removed by eigenvector overlap) give **133 residual-certified eigenpairs** in $[1.75705, 2.11657]\,\mu\text{m}^{-2}$ (worst residual 5.9×10^{-5} in fp64, 8.7×10^{-5} as reported by the solver; 104 GPU-hours, 1.16×10^7 operator applications), with no in-window pair left unconverged. Completeness is certified on the sub-interval $[1.9063, 1.9606]$ by a deflated-probe KPM count of missed states, $|\text{missed}| < 0.01$ (stochastic s.e. 10^{-4}; the estimator's own floor from modelled leakage and eigenvector error is $\sim 3 \times 10^{-3}$, against a signal of ≈ 1 per missed state); on the full window the same estimator ($+1.7 \pm 0.3$) carries an edge-leakage bias $\geq 1.2 \pm 0.2$ and is consistent with zero or one missing state (SM). The largest interior spacing of the certified spectrum runs from 1.8860 to $1.9264\,\mu\text{m}^{-2}$ ($\Delta\nu/\nu = 1.06\%$, 28× the median spacing; completeness is certified only above 1.9063); but the nominal gap is not empty. **Ten certified states lie inside** $[1.864, 1.996]$, at $\lambda = 1.8690$, 1.8708, 1.8731, 1.8860, 1.9264, 1.9296, 1.9441, 1.9472, 1.9738, 1.9901 [Fig. 1(d), Table SI]. Four of them are flagged as boundary-seam states of the rasterization convention: rods whose radius pokes through a box face are not wrapped, so the outermost voxel shell has ff = 0.1975 against 0.2211 inside (an 11% deficit, a planar defect), and the states at 1.8708, 1.8731, 1.9296, 1.9472 carry 18, 24, 44 and 42% of their energy in that shell (6.1% of the volume; three peak on a box edge). Re-solving the gap window with minimum-image wrapping (0.078% of voxels change) finds no partner for three of them (best overlaps 0.006, 0.131, 0.090), but the periodic solve is itself measured incomplete ($+2.0 \pm 0.4$ missed), so their absence is likely, not proven; the fourth (1.9472, overlap 0.30) is undetermined. A fifth in-gap state (1.9441) is compact ($p = 0.56\%$) but has a non-exponential envelope ($r^2 = 0.32$, ξ unresolved); its periodic partner (overlap 0.95) fits as $\xi = 2.2\,\mu$m, and it is excluded from the candidate list conservatively. **Five bulk-localized in-gap states remain** among the certified set: ξ = 1.80–2.51 μm (ξ/L = 0.07–0.10), p = 0.034–0.33% (participation volumes 5–49 μm^3 in a 15 000 μm^3 box), r^2 = 0.971–0.998, shell energy 2.5–10.5%; all five persist under periodic wrapping with overlaps 0.968–1.000 and shifts $\Delta\lambda = -0.0007$ to -0.0030, every shift negative as required when dielectric is added. A 2.4× voxel refinement (256^3) on two anchor windows matched 11 of 11 states in $[1.84, 1.95]$ one-to-one by eigenvector overlap (min 0.873, median 0.996, the two lowest a rotating near-degenerate pair; $|\Delta\omega/\omega| \leq 0.34\%$) and 10 of 11 in $[1.99, 2.035]$ ($\leq 0.04\%$; but only 3 of 11 fine pairs are certified and the rest carry Weyl uncertainties up to 0.10%, so the high-edge shift is not resolved below that). Three candidates (1.8690, 1.8860, 1.9264) lie in the low anchor and persist; 1.9738 was not tested, and 1.9901, at the high-anchor floor, has no fine partner. This is good evidence against a rasterization artefact for the tested states; with 7.8 voxels/μm and the 0.27% spread seen at $N = 1000$, sub-percent gap-edge frequencies are not claimed.

The decay length varies smoothly across the window [Fig. 3(a)]: 121 of 133 fits are resolved (12 fail $r^2 \geq 0.7$), and the medians by spectral bin are $5.0 \to 3.0 \to 2.1\,\mu$m approaching the lower edge, $2.1\,\mu$m inside, and $2.8 \to 4.7 \to 6.1\,\mu$m climbing away above; p spans 0.03–12% [Fig. 3(b)], and the participation-volume radius $(3V_p/4\pi)^{1/3}$ follows the same funnel ($2.7 \to 1.5 \to 1.2\,\mu$m and $1.4 \to 3.1 \to 4.1 \to 5.4\,\mu$m). The envelope fit is not a precision quantity: shrinking the fit range to $[0.10, 0.60]\,L/2$ moves ξ by a median 30% and the window-end/near-edge contrast from 1.7–2.2× to 1.6–1.8×; the funnel is robust, three-figure ξ values are not.

Can we claim Anderson localization?—We take the items in turn. (1) *Spatial localization*: supported. The five bulk in-gap candidates have ξ = 1.8–2.5 μm in $L = 24.6\,\mu$m, $p \leq 0.33\%$ and $r^2 \geq 0.97$ [Fig. 2(b)]. ξ is an envelope-fit length with tens-of-percent fit-range sensitivity, not a transport length. (2) *Interference-driven (Anderson) vs trap-like (Lifshitz-tail [31]) states*: not separable with this data. Both boxes are single realizations with the same local statistics (ff coarse-grained at 2 μm: 0.2204 vs 0.2209, s.d. 0.0398 vs 0.0405) but $N = 10^4$

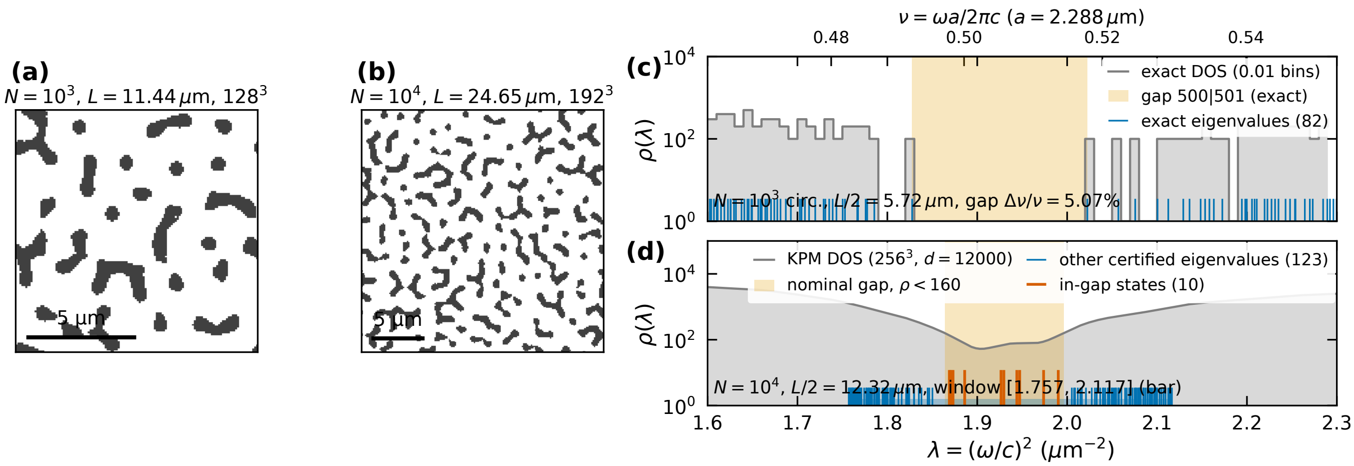

FIG. 1. (a,b) Mid-plane $\varepsilon(\boldsymbol{r})$ of the $N = 10^3$ (128^3) and $N = 10^4$ (192^3) networks, circular decoration (dark = dielectric). (c) $N = 10^3$, circular decoration: exact eigenvalues (bottom-up, 611 bands) around the gap and their binned density of states; the shaded band is the exact gap between MPB bands 500 and 501. (d) $N = 10^4$: KPM density of states (256^3) with the 133 residual-certified 192^3 eigenvalues (blue) and the ten inside the nominal gap $\rho < 160$ states per unit λ (red); bar: solved window. Top axis: $\nu = \omega a/2\pi c$, $a = 2.288\,\mu$m.

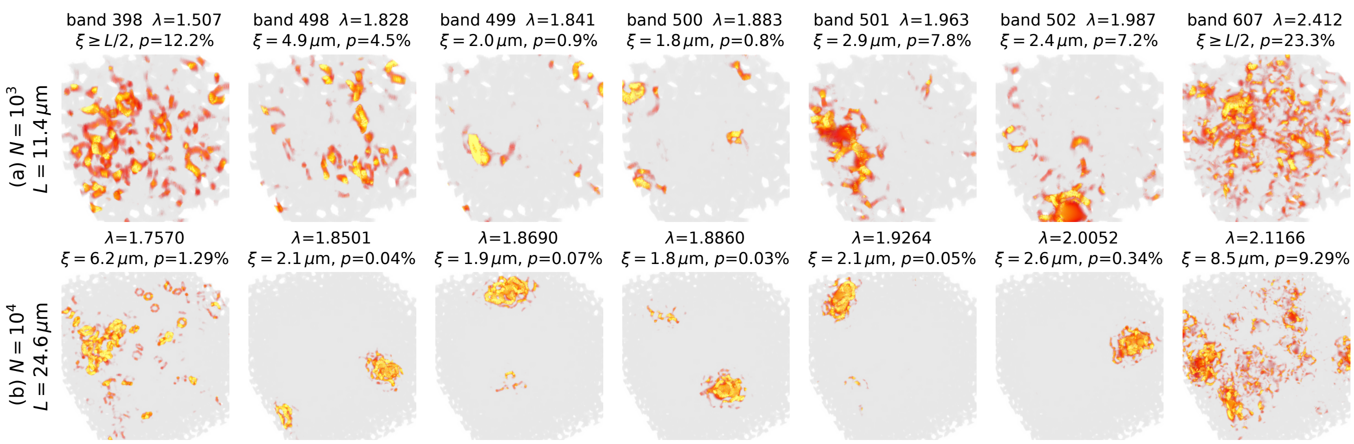

FIG. 2. Energy density $\varepsilon|\boldsymbol{E}|^2$ (volume render, per-tile colour scale clipped at the 99.9th percentile; network in translucent grey). (a) $N = 10^3$, elliptical decoration, bands 398, 498–502, 607 around the 500|501 gap. (b) $N = 10^4$, circular decoration: window bottom (1.7570), the nearest state below the nominal gap (1.8501), three of the five in-gap candidates (1.9738 and 1.9901 not shown), the nearest state above it (2.0052), window top (2.1166). ξ and p as defined in the text; "$\xi \geq L/2$" marks a ceiling-limited fit.

has ten times as many ξ-cells; at 5 in-gap states per $N = 10^4$ volume an empty $N = 1000$ gap has probability $e^{-0.5} = 0.61$—but over the interval $[1.8276, 2.0225]$ where the $N = 1000$ gap is actually empty the $N = 10^4$ spectrum holds 19–20 non-seam states, giving $e^{-2} \approx 0.14$, so the statistics argument is plausible, not compelling. A direct test (energy-weighted local filling fraction of candidates vs bulk controls) was run and *retracted*: unmatched in mode extent, it reverses against an extent-matched null (SM). Neither origin is established. (3) *Finite-size scaling*: two sizes only. Matched-decoration band-edge states have $\xi = 1.47$ and $2.30\,\mu$m at $L = 11.44\,\mu$m and 1.80 and $2.09\,\mu$m at $L = 24.65\,\mu$m [edge states 1.8860 and 1.9264; Fig. 4(b)], with participation volumes 6 and $58\,\mu\text{m}^3$ vs 5 and $7\,\mu\text{m}^3$: ξ is L-independent within the fit uncertainty, but at the estimator's 30% systematic this two-point test has little discriminating power (the upper-edge participation volumes differ by 8×), two points are not a scaling analysis, and 168 of 210 $N = 1000$ fits are unresolved (142 at the ceiling). (4) *Level statistics* [32, 33]: Θ at Γ with real ε is real symmetric, so the class is orthogonal: $\langle r\rangle_{\text{P}} = 2\ln 2 - 1 = 0.386$, $\langle r\rangle_{\text{GOE}} = 0.531$ [34, 35]. The adjacent-gap ratio (no unfolding needed) is computed within contiguous bands [Fig. 4(a)]. In the extended regions it is GOE-like: 0.528 ± 0.038 (51 levels, $\lambda < 1.816$) and 0.528 ± 0.041 (42 levels, $\lambda > 2.05$) at $N = 10^4$, pooled 0.519 ± 0.024 outside the bracket; 0.508 ± 0.010 and 0.533 ± 0.011 over

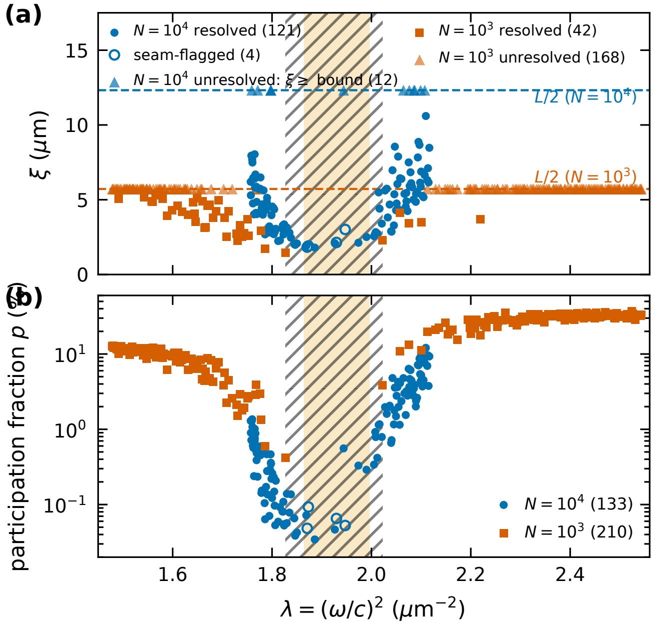

FIG. 3. (a) Envelope decay length ξ vs λ for $N = 10^4$ (circles, 192^3) and $N = 10^3$ circular decoration (squares, 128^3); dashed lines are the finite-size ceilings $L/2$; triangles at the ceiling are unresolved fits (lower bounds); open circles are the four seam-flagged states; shaded: nominal $N = 10^4$ gap; hatched: $N = 10^3$ gap. (b) Participation fraction. Same decoration for both sizes.

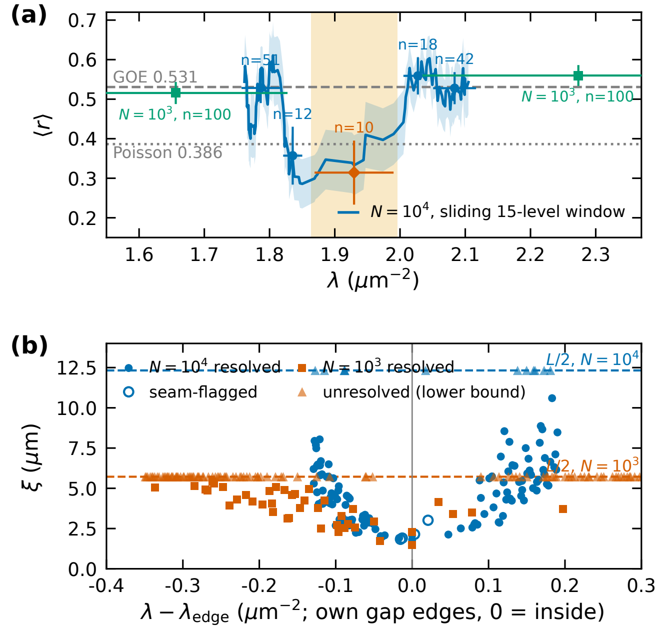

FIG. 4. (a) Adjacent-gap ratio $\langle r\rangle$: $N = 10^4$ sliding 15-level window (line, ±1 s.e.; inside the bracket it includes the seam-flagged levels and ratios dominated by the two largest spacings, and is not interpreted) and contiguous bands (circles: 51, 12, 18, 42 levels; diamond: ten in-gap levels; bootstrap errors); $N = 10^3$ circular, bands 401–500 and 501–600 (squares). Dashed/dotted: GOE 0.531 / Poisson 0.386. (b) Matched-decoration comparison: resolved ξ vs distance from each box's own gap edges ($N = 10^3$: 1.8276|2.0225; $N = 10^4$: 1.8860|1.9264; in-gap states at 0).

all 611 $N = 1000$ levels (elliptical, circular). In the 12 levels just below the gap ($1.8208 \leq \lambda \leq 1.8502$, a post-hoc window), where $\xi = 2.0$–$3.4\,\mu$m, $\langle r\rangle = 0.357 \pm 0.073$, 2.2σ below the GOE expectation for 12 levels and consistent with Poisson; four of these 12 levels carry outer-shell enhancement $> 2\times$ (up to 25%, comparable to the seam-flagged in-gap states) and are not excluded. Above the gap the 18 nearest levels (2.005–2.050, $p \approx 1\%$, ξ median 4.1 μm) give 0.559 ± 0.047: comparably compact states show repulsion there, an unexplained asymmetry. The ten in-gap levels give 0.31 ± 0.08, but ten levels, four of them seam-flagged, are not a statistic. The residual gate guarantees $|\Delta\lambda| \leq 10^{-4}\lambda \approx 1.9 \times 10^{-4}\,\mu$m^{-2} per level (Weyl), 14% of the median spacing $1.4 \times 10^{-3}\,\mu$m^{-2}, while the accuracy measured on the $N = 1000$ ground truth is $\Delta\lambda/\lambda \leq 4 \times 10^{-7}$ and duplicate states found by two slices agree to 10^{-8}; a single missing level would move the 12-level ratio by up to ~ 0.1. Level repulsion where states are extended and its loss where they are localized is the expected pattern, but the near-edge samples are small, post hoc and seam-contaminated: suggestive, not conclusive. A Thouless-type sensitivity to boundary twists cannot be obtained from the saved Γ-point data. (5) *Mobility edge*: not claimable—no transport, one realization per size, no scaling; the $\xi(\omega)$ funnel (Fig. 3) and the $\langle r\rangle$ drop [Fig. 4(a)] are the *phenomenology* of a localization–delocalization crossover at the gap edge, the regime probed experimentally in 3-D amorphous networks [14] and predicted for correlated disorder [13]. (6) *Vector-wave caveat*: early reports of 3-D localization of light [36] were not confirmed, vector waves do not localize in uncorrelated point-scatterer ensembles [37–39], and numerical 3-D localization has been shown only for overlapping metallic, not dielectric, spheres [40]; localization at the edge of a photonic gap in a correlated network [12, 13, 41] is a different mechanism; this data addresses only the spatial structure of the eigenstates at such an edge.

Verdict.—The defensible claim is: strongly spatially localized in-gap and gap-edge eigenstates with $\xi \ll L$, with the two resolved $N = 1000$ band-edge values in the same range, consistent with localization at the edges of a photonic pseudogap in a correlated disordered network, accompanied by GOE-like level statistics in the extended bands and a Poisson-compatible ratio in 12 post-hoc levels at the lower edge that is not reproduced above the gap. Whether these states are interference-driven (Anderson-type) or trapped in rare local configurations (Lifshitz-tail-type) cannot be decided from this data. The claim of an Anderson transition or a mobility edge requires finite-size scaling over three or more sizes and/or transport or larger-sample level statistics that this data does not contain.

Limitations.—Gap-edge frequencies are rasterization-limited (0.27% non-monotone spread at $N = 1000$; $\leq 0.34\%$ between 192^3 and 256^3 at $N = 10^4$), and on MPB's interpolated read-back of the 64^3 grid the elliptical 500|501 spacing is 0.60% (largest at 501|502), so the gap position at 64^3 is convention-dependent. Decay lengths are bounded by $L/2$ and fit-range dependent at the 30% level; one disorder realization per size. Arithmetic is fp32 on the device with fp64 Rayleigh–Ritz; the merged 133-mode set is orthonormal only to 4.9×10^{-4} across independently solved slices (8.6×10^{-6} within), the level the 10^{-4} residual gate permits, which affects no spectrum, montage or localization result. Twenty of the 133 modes have outer-shell enhancement $> 2\times$; only the four inside the bracket were re-solved seam-free. Of the statements withdrawn during the project (eight are listed in the SM), four bear on the results here and are absent: the rare-region z-test, the statement that the resolution confound is "closed" (its power-retention statistic is saturated), the "vanished" verdict for the state at 1.9472, and all uncorrected eigenvalues. Full records and every number with its source file are in the SM [30].

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